# IT Ticket Classification for Application Management Services using Classical Machine Learning Techniques*

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**Abstract**

IT outsourcing company is widely used for many organizations to support applications used in the organizations without interrupting business processes. The challenging problem for IT outsourcing company is that a helpdesk officer must specify the reported problem for which supported applications, and the officer must forward the ticket to a correct expert to solve that problem. If many errors occur, the company could breach the Service Level Agreement (SLA). This paper proposed a framework to handle imbalanced dataset with Thai and English languages and to use 8 machine learning models, i.e. Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Logistic Regression (LR), Multinomial Naïve Bayes (MNB), Decision Tree (DT), Bernoulli Naïve Bayes (BNB), Light Gradient-Boosting Machine (LightGBM), and Extreme Gradient Boosting (XGB) to classify IT tickets since the models required less computational resources when comparing to deep learning models. The dataset is from ServiceNow application in an IT outsourcing company with 20,586 records to be classified into 142 applications, which is highly imbalanced dataset. The performance metrics are accuracy, precision, recall, and F1-score. From two experiments using a class weight with lemmatization, and 6 input features, SVM model gives the best F1-score at 85.71% and the features are short description, description, business unit, sub business unit, and company code.

* Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. 34, July 8–10, 2026, Seoul, South Korea.
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# 1 Introduction

In the modern business process, a large organization utilizes Information Technology (IT) in managing business tasks but IT knowledge is not for everyone. Therefore, the organization can outsource IT supports to an IT outsourcing company to help business handle technology-related functions. Application Managed Services (AMS) play an important role for business continuity to be responsible for many applications. Each application may be complex and required an expert in solving a problem.

From this context, IT Service Management (ITSM) system, e.g. ServiceNow, is used for recording, tracking, and managing cases or incidents from several IT systems in organizations. When a user finds a problem or needs technical assistance, the user contacts a Helpdesk officer, and reported information will be recorded in the form of a case or a ticket in ServiceNow. Then, the Helpdesk officer will assign the case to an expert for each application to solve the problem later.

For the case assignment, a user reports a problem to a Helpdesk officer using a phone, an email, or a self-service portal in ServiceNow. The reported information, called a ticket, i.e. short description, description, category, priority/urgency, application/service, and caller information, will be recorded for each case. After getting the ticket, the Helpdesk officer must analyze the case information and assign the case to an expert who is responsible for solving that problem directly. The process of the case assignment needs a deep knowledge in the scope of each application but there can be more than hundred supported applications.

The ServiceNow application can manage the workflow efficiently, but the case assignment still needs to do it manually and this causes several problems as follows. The first problem is the misclassification. An officer assigns a newly received ticket to incorrect experts. The second problem is the delay in case escalation. After the misclassification process, the wrongly assigned expert must reassign to a correct expert. This causes a delay in solving the case and the response time or resolution time, which is one of the performance indexes in Service Level Agreement (SLA), is increased. From the delay, the IT outsourcing company can get the SLA breach penalty, and the Customer Satisfaction Score (CSAT) can be reduced in the long run. The third problem is the knowledge dependency. Since the case assignment task needs an officer to have a lot of experience but when experienced officers quit the jobs or transfer to other companies, the knowledge is not transferred properly. The case assignment quality is lower during the transient period. The last problem is multilingual data challenge. For Thai IT outsourcing company, case information is recorded in both Thai and English languages. The case classification is more complex for the officers.

From the problems mentioned above, several researchers proposed to use machine learning and natural language processing techniques to apply for IT Helpdesk ticket classification automatically. It means that the use of machine instead of human to classify IT tickets is needed to reduce human errors and the misclassification delay. However, high number of application classes (more than 100 applications) and the imbalance class are a challenge research topic. Therefore, this paper proposed to use machine learning techniques for IT ticket classification with the high number of application classes and class imbalance problems. The eight classical machine learning techniques in this research are Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Logistic Regression (LR), Multinomial Naïve Bayes (MNB), Decision Tree (DT), Bernoulli Naïve Bayes (BNB), Light Gradient-Boosting Machine (LightGBM), and Extreme Gradient Boosting (XGB). The models require less computational resources than deep learning models. In addition, they are suitable for AMS services in terms of speed, operation costs, and maintenance. The contributions of this paper are proposing a related application feature design using hybrid inputs to the models, using text tokenization and lemmatization techniques instead of stemmer for Thai and English languages, and proposing the imbalance techniques for a big, highly imbalance, and practical dataset.

The organization of this paper is as follows. The next section is related work on the IT ticket classification. Then, the proposed work is explained with the framework, data collection, data preprocessing, input feature selection, and machine learning models. Next, proposed experiments are described in detail with experimental setup, and performance metrics. Then, experiment results and analysis section is discussed and analyzed on the results. The last section is a conclusion and future work to summarize this paper and offer possible future experiments.

# 2 Related work

The research in the IT ticket classification has been widely proposed using machine learning, and deep learning techniques. For machine learning technique, Paramesh et al. (2018) proposed to use 5,550 IT service desk records to classify into 12 classes with English language only. The algorithms in the study were Multinomial Naïve Bayes (MNB), Support Vector Machine (SVM), Decision Tree (DT), Random Forest, and K-Nearest Neighbor (KNN) together with Ensemble Bagging technique. The feature representation was Term Frequency-Inverse Document Frequency (TF-IDF) and the class imbalance problem was solved using Random Oversampling method. The best F1-score was 60.76% for SVM technique. Parmar et al. (2018) proposed to use a bigger dataset with 10,742 records in 18 classes using Voting classifier mixed with Logistic Regression, KNN, MNB, SVM, and Random Forest techniques. The features were description and category with TF-IDF transformation. The highest F1-score was 88.14% using SVM. Al-Hawari and Barham (2019) proposed to use title and description as features. The best machine learning technique was SVM with F1-score at 81.4%. The main problems for this research were that the size of dataset was small and the classes were not more than 18 classes. They cannot reflect the complexity in large, practical IT outsourcing companies for IT ticket classification.

For multilingual IT ticket problem especially Thai and English languages, Pidej and Thaicharoen (2021) showed that using the same dataset and machine learning models in their study, the accuracies for Thai and English only were 64.93% and 86.21%, respectively, and the best F1-score was 87.81%. There was a huge difference 21% for the accuracy with different language. Sungthong et al. (2024) proposed to use PhayaThaiBERT model, which was pre-trained with Thai language specifically. The F1-score was 76.9% with 940 records for the dataset. For Thai language IT Tickets, Khowongprasoed and Titijaroonroj (2022) proposed to use IT Infrastructure Company dataset in Thailand with 360 records. In this study, there were three classes, i.e. incident, request, and problem. SVM technique gave the best F1-score at 72%. The multilingual environment is another challenge for IT ticket classification problem. Table 1 shows the summary table for related work.

| Reference | Technique | Data size | Number of classes | Language | Model | F1-score |
|---|---|---|---|---|---|---|
| Paramesh et al. (2018) | SVM, NB, DT | 5,550 | 12 | ENG | SVM | 60.76% |
| Parmar et al. (2018) | Voting SVM | 10,742 | 18 | ENG | SVM | 88.14% |
| Al-Hawari and Barham (2019) | SVM, NB, DT | 1,254 | 13 | ENG | SVM | 81.40% |
| Pidej and Thaicharoen (2021) | mBERT, ULMFiT, XLM-R | 11,572 | 14 | TH/ENG | XLM-R | 87.81% |
| Sungthong et al. (2024) | PhayaThaiBERT | 940 | 30 | TH/ENG | Deep Neural Network | 76.9% |
| Khowongprasoed and Titijaroonroj (2022) | SVM | 360 | 3 | TH | SVM | 72.0% |

**Table 1: Related work summary**

From the literature work, the dataset size is not more than 12,000 records and the number of classes is not more than 30 classes. In addition, most research used title or short description as an input to machine learning models. Therefore, this research proposed a framework to handle Thai and English languages and to explore classical machine learning techniques to classify IT tickets with highly imbalance classes in a large dataset with multilingual environment using several input features.

# 3 Proposed work

This section is to explain the proposed work starting with proposed framework to give the overview of this work. Then, data collection, data preprocessing, input feature selection, and machine learning models used in this paper are described in detail.

## 3.1 Proposed framework

The framework of IT ticket classification is shown in Figure 1. The data collection is to collect the IT ticket data for experiments. The data preprocessing is to clean and prepare the data before feeding into machine learning models. The last part is to evaluate and analyze the results. The explanations for each part are as follows.

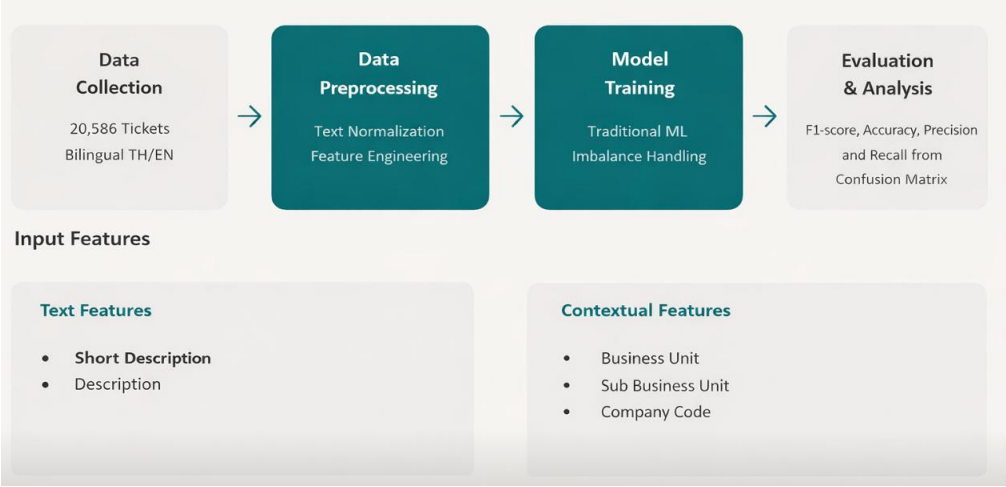



**Figure 1: IT ticket classification framework**

## 3.2 Data collection

The dataset has been collected for 1 year from IT outsourcing company in Thailand and total IT tickets in ServiceNow application was 20,586 records. The IT tickets were opened in the system from emails, self-service portal, and phones in both Thai and English languages to be classified into 142 applications. Duplicate or canceled tickets were removed from the dataset. The dataset is highly imbalanced as shown in Figure 2. Some applications are reported more frequently than other applications. This dataset consists of 6 input features and one output feature as follows.

1. *Short description* is a short problem inputted from a user and a helpdesk officer.
2. *Description* is a detailed problem or error messages.

3. *Business unit* is a main business unit of a user.
4. *Sub business unit* is a sub business unit of a user.
5. *Company code* is a code of a company of a user.
6. *Contact type* is the communication channel, i.e. email, portal, call center, to open a ticket.
7. *Service offering* is the name of applications, which is a target variable to be classified and used for supervised learning.

The example dataset is shown in Figure 3, which contains both Thai and English languages and some technical terms in the description. However, the service offering field is not shown.

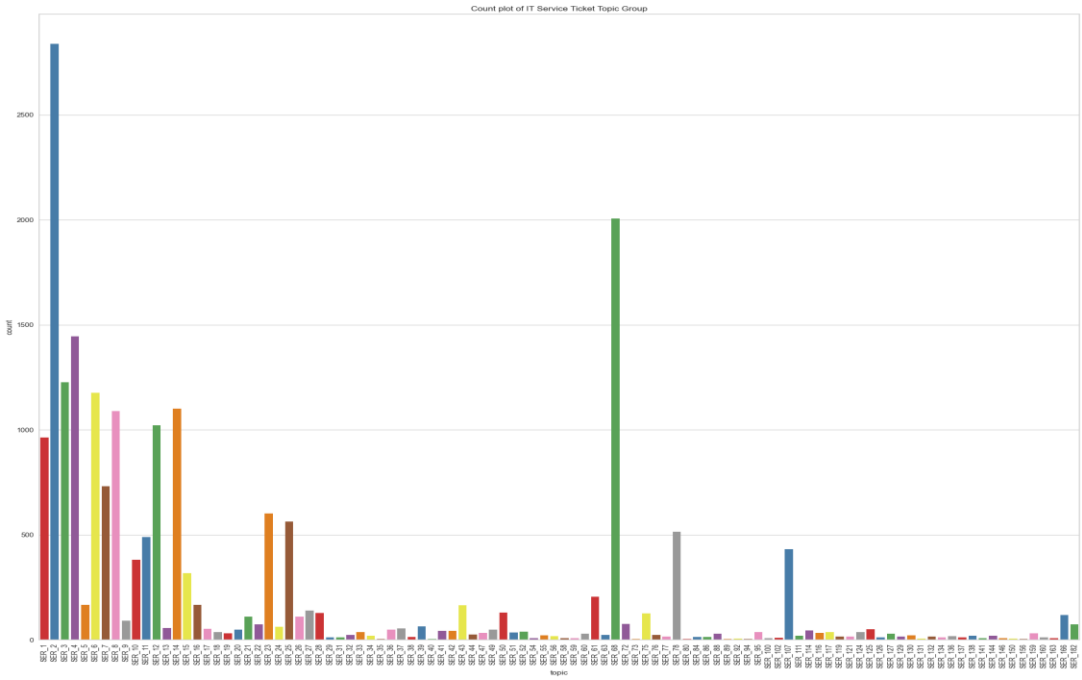



**Figure 2: Application distribution in the dataset**

| Business | Sub | Company | Service offering | Short description | Description | Contact |
|---|---|---|---|---|---|---|
| BUS_3 | | COM_3 | SER_7 | ตรวจสอบ HYYYYB Lot YHYYYYYY Reverse ไม่ได้ | ตรวจสอบ HYYYYB Lot YHYYYYYY Reverse ไม่ได้<br>รายละเอียดเพิ่มเติมตามไฟล์แนบ | Email |
| BUS_4 | SUB_2 | COM_4 | SER_1 | ตรวจสอบ ใบตัดจ่าย ไม่มีการ run billing SAP expo เมื่อคืนนี้ทำให้วันนี้ไม่มียอด | ตรวจสอบ ใบตัดจ่าย ไม่มีการ run billing SAP expo เมื่อคืนนี้ทำให้วันนี้ไม่มียอด<br>ซึ่งตอนนี้พบมา Y ร้านแล้ว<br>ตัวอย่าง DN YYYYYYYYY | Phone |
| BUS_4 | SUB_3 | COM_5 | SER_8 | ตรวจสอบการบันทึกเวลาของพนักงานขาดงาน Y ชม. (YY.YY-YY.YY,YY.YY-YY.YY) แต่ระบบขึ้นมาสาย Y ชม.ขาดงาน Y ชม. | ตรวจสอบการบันทึกเวลาของพนักงานขาดงาน Y ชม.(YY.YY-YY.YY,YY.YY-YY.YY) แต่ระบบขึ้นมาสาย Y ชม.ขาดงาน Y ชม. | Email |
| BUS_1 | SUB_1 | COM_1 | SER_1 | [HANA Retail] ตรวจสอบ IYY - เมื่อยิง mode fix พบว่าไม่สามารถสร้าง DN ต่อได้ | ตรวจสอบ เมื่อ ยิง IYY แล้วดิด error <MESSAGE>No ATP for target delivery qty</ MESSAGE> ไม่สามารถสร้าง DN ต่อได้<br><br><PIID_OUT>YYYYYYYBEBYFYYEACYYYYYYYYYYYYYYA</PIID_OUT><br><PIID>YYYYYADYDAFDYYEABYEYYYYYYYYYYYBF</PIID><br><FC_PO>AYYYYYYYYYYYY</FC_PO> | Email |
| BUS_3 | | COM_6 | SER_9 | ตรวจสอบไม่สามารถเข้าไปแก้ไข Elements ต่างๆ ใน PI AF ได้ | ตรวจสอบไม่สามารถเข้าไปแก้ไข Elements ต่างๆ ใน PI AF ได้ ซึ่งปกติก่อนหน้านี้จะสามารถเข้าไปแก้ไข Element ต่อไปนี้ได้<br><br>Y. VCM Process Monitoring<br>Y. VCMY<br>Y. VCMY<br>Y. Envi Monitoring | Portal |

**Figure 3: Example dataset**

## 3.3 Data preprocessing

Before feeding the data to a classification process or machine learning models, the raw data must be pre-processed systematically to clean the IT ticket data as follows.

1. Decommissioned application removal

Tickets related to decommissioned applications were removed from the dataset because they are not supported in the ServiceNow application.

2. Non-Thai/English character removal

Some IT tickets consist of special/symbol characters, non-Thai/English characters including HTML tags, URL, reference numbers, and non-printable characters. These characters are removed from short description and description fields in order to reduce noise in the dataset.

3. Stop word removal

All stop words from both Thai and English languages are removed from the dataset because these words give high frequency without statistical significance on IT ticket classification. This removal can increase the performance of machine learning models.

4. Lemmatization

Lemmatization is a technique for transforming words into a lemma or a base form to reduce the complexity and increase the quality of feature representations. It helps the models to classify IT tickets correctly.

For English language, NLTK library was used for text cleansing and text normalization, for example, converting upper case to lower case, removing punctuations, and lemmatization, in order to convert words back into basic words, which can reduce the word variations and increase the performance of the machine learning models. Then, the tokenization and stop word removal were performed before feeding into the models.

For Thai language, PyThaiNLP library and NewMM algorithm were used for tokenization by referring to words in Thai dictionary, which was designed for Thai language processing only. Then, the standard stop word in PyThaiNLP library was also used for stop word removal.

## 3.4 Input feature selection

From the dataset, the data can be classified into 2 groups, i.e. textual data and organizational context data. The textual data is data related to a problem, which are short description and description. The organizational context data is data related to an organization that reports a problem, which are business unit, sub business unit, and company code. Some applications are used in specific companies, and these data could enhance the classification performance. In the experiments, the input features are incrementally added to machine learning models as follows.

1. Baseline 1: Short description only
2. Hybrid 2: Short description, Description
3. Hybrid 3: Short description, Description, Business unit
4. Hybrid 4: Short description, Description, Business unit, Sub business unit
5. Hybrid 5: Short description, Description, Business unit, Sub business unit, Company code

Figure 4 shows all hybrid features in this experiment to help understand the effects of each feature to enhance the classification performance.

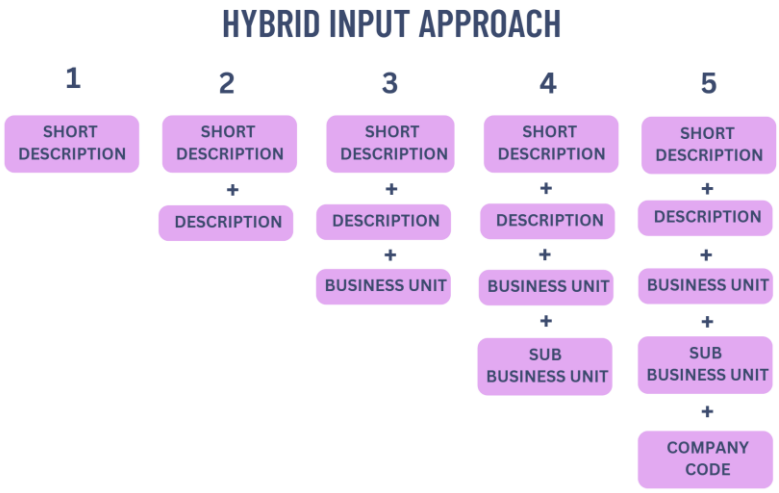



**Figure 4: Input features**

## 3.5 Machine learning models

From the related work, several classical machine learning models were implemented and tested with different performance metrics. Classical machine learning models require less computational resources and memory than deep learning models. In this paper, eight classical machine learning models are selected from supervised learning models as follows.

1. Support Vector Machine (SVM) is the algorithm to find an optimal hyperplane to classify class in high dimensional feature space using linear kernel with one-vs-rest strategy for multi-class classification.
2. Logistic Regression (LR) is a linear classification algorithm using Sigmoid function to estimate a probability of each class with regularization to avoid an overfitting problem.
3. Multinomial Naïve Bayes (MNB) is a probabilistic algorithm for the data transforming to term frequency using the conditional independent assumption between features.
4. Bernoulli Naïve Bayes (BNB) is a Naïve Bayes algorithm using binary feature representation. It is good for appearance or absence of terms.
5. Stochastic Gradient Descent (SGD) is a linear classification algorithm using stochastic updated weights and good for a large dataset.
6. Decision Tree (DT) is a tree-like structure algorithm to select features with maximum information gain for each level.
7. Extreme Gradient Boosting (XGB) is an ensemble algorithm using gradient boosting that creates several decision trees sequentially. Each tree is created by learning errors from the previous tree.
8. Light Gradient-Boosting Machine (LightGBM) is a gradient boosting algorithm using leaf-wise growth and histogram-based splitting to enhance the training in a large dataset.

## 3.6 Feature representation

The textual data is transformed into numerical feature vector using Term Frequency-Inverse Document Frequency (TF-IDF) that gives a weight to each word depending on frequency of the appearance in a document and reduces a weight for a word that appears in many documents to balance the overall weights in the feature. TF-IDF (min_df=2, ngram_range=(1,2)) was used for transforming texts to vectors by considering the weights from keywords. Hyperparameters were set to Regularization Parameter =1, Tolerance=0.0001.

## 3.7 Class imbalance handling

The dataset mentioned in the previous section is highly imbalanced that some applications had high number of IT tickets, but some applications has very low number of tickets. In this paper, two techniques are studied to solve this problem as follows.

1. Random oversampling (ROS) is a technique to increase samples for minority classes by randomly copying minority data classes to roughly match the majority classes.
2. Class weight (CW) is a technique to adjust weights in a loss function by giving high weights to minority classes such that the models give the priority to minority classes. The weight is inversely proportional to number of samples for each class.

# 4 Experiments

## 4.1 Experiments setup

From the study, two sets of experiments were performed. The former set, called Experiment 1: Finding the best class imbalance technique, is to find the best technique (random oversampling or class weight) for class imbalance by using a short description as an input feature with stemming. The latter set, called Experiment 2: Finding the best input features, is to fine the best input features for all machine learning models using the best class imbalance technique for the former set. The lemmatization is used instead of stemming in this set.

The dataset consists of 20,586 IT Ticket records and splits into a training set and a test set with 70% and 30% respectively using Stratified split for dealing with class imbalance to keep the balance of the ratio of service offerings between the training set and the test set, instead of chronological split. However, before using the stratified split, the class distribution was under investigated and if a class has less than 2 samples, the class will be removed for preventing the errors, since stratified split requires to have a sample in both training and testing sets.

## 4.2 Performance metrics

In this study, there are 4 performance metrics, i.e. accuracy, precision, recall, and F1-score. However, these metrics use a confusion matrix as a background, and it is explained as follow.

*Confusion matrix* is a table to evaluate the performance of a classification machine learning model by comparing predicted classes with actual classes as shown in Table 2.

| | **Actual positive** | **Actual negative** |
|---|---|---|
| **Predicted positive** | True positive (TP) | False positive (FP) |
| **Predicted negative** | False negative (FN) | True negative (TN) |

**Table 2: Confusion matrix**

1. *Accuracy* is a ratio of accurately classified IT Tickets over total number of IT Tickets, as shown in (1).

$$accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (1)$$

2. *Precision* is a ratio of classified IT Tickets as positive over total number of both true positive and false positive actual IT Tickets, as shown in (2).

$$precision = \frac{TP}{TP+FP} \quad (2)$$

3. *Recall* is a ratio of classified IT Tickets as positive over total number of both true positive and false negative actual IT Tickets, as shown in (3).

$$recall = \frac{TP}{TP+FN} \quad (3)$$

4. *F1-score* is a harmonic mean of precision and recall as shown in (4), which is a main performance metric in this paper because it is good for imbalanced dataset.

$$F1 - score = \frac{2 \times precision \times recall}{precision + recall} \quad (4)$$

For multi-class classification, there are 142 classes for this dataset, and the accuracy, precision, recall, and F1-score were calculated as weighted average to reflect the overall performance of the models for each class to reduce a variation of the performance from small sample classes.

# 5 Experiment results and analysis

## 5.1 Experiment 1: Finding the best class imbalance technique

In this experiment, two class imbalance techniques, which are a random over sampling technique and a class weight technique, are under investigated. Figure 5 shows the results from a short description input with the random over sampling technique to overcome the class imbalance problem. The results show that Linear Support Vector Classification (LinearSVC) or Support Vector Machine (SVM) gives the highest F1-score at 79.12%. Then, the results from a class weight technique instead of random over sampling technique, are shown in Figure 6. The highest F1-score is 79.74% for the SVM model. When lemmatization is used instead of text stemming shown in Figure 7, the F1-score is increased to 79.77% for SVM models. Figure 8 shows the summary of finding the best class imbalance technique by using SVM with the class weight and lemmatization techniques, and the highest F1-score is 79.77%.

The most misclassification application is Application SER_114, which has 13 records. F1-score and recall for this application were 13% and 15% respectively. This application was new and had same functions with other applications but different user interface (UI) in the same Business Unit (BU) and Company. However, different applications for different factory can have the same functions in the same BU. This is a difficult problem for classification.

```
Feature Input     : Short description
Total Classes     : 142
BASE_MODEL        : True
LEMMARTIZING      : False
IMBALANCE MODE    : ros
CALIBRATED_MODE   : False
TUNED MODE        : False
```

| Model | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| LinearSVC + ROS | 0.7847 | 0.8157 | 0.7847 | 0.7912 |
| Logistic Regression + ROS | 0.7820 | 0.8094 | 0.7820 | 0.7880 |
| MultinomialNB + ROS | 0.7498 | 0.8309 | 0.7498 | 0.7778 |
| BernoulliNB + ROS | 0.7086 | 0.8170 | 0.7086 | 0.7480 |
| XGB + ROS | 0.7117 | 0.7624 | 0.7117 | 0.7205 |
| SGDClassifier + ROS | 0.6579 | 0.8591 | 0.6579 | 0.7194 |
| LightGBM + ROS | 0.7054 | 0.7380 | 0.7054 | 0.7079 |
| DecisionTree + ROS | 0.6892 | 0.7134 | 0.6892 | 0.6943 |

**Figure 5: Results from a short description with random over sampling technique**

```
Feature Input     : Short description
Total Classes     : 142
BASE_MODEL        : True
LEMMARTIZING      : False
IMBALANCE MODE    : class_weight
CALIBRATED_MODE   : False
TUNED MODE        : False
```

| Model | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| LinearSVC + classweight | 0.7813 | 0.8356 | 0.7813 | 0.7974 |
| SGDClassifier + classweight | 0.6935 | 0.8457 | 0.6935 | 0.7412 |
| Logistic Regression + classweight | 0.6851 | 0.8329 | 0.6851 | 0.7353 |
| MultinomialNB + SampleWeight | 0.6382 | 0.8782 | 0.6382 | 0.6981 |
| DecisionTree + classweight | 0.6814 | 0.7106 | 0.6814 | 0.6898 |
| BernoulliNB + SampleWeight | 0.5482 | 0.7943 | 0.5482 | 0.6001 |
| LightGBM + classweight | 0.5175 | 0.7307 | 0.5175 | 0.5869 |
| XGB + SampleWeight | 0.4408 | 0.6934 | 0.4408 | 0.5140 |

**Figure 6: Results from a short description with class weight technique**

```
Feature Input     : Short description
Total Classes     : 142
BASE_MODEL        : False
LEMMARTIZING      : True
IMBALANCE MODE    : class_weight
CALIBRATED_MODE   : False
TUNED MODE        : False
```

| Model | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| LinearSVC + classweight | 0.7818 | 0.8356 | 0.7818 | 0.7977 |
| SGDClassifier + classweight | 0.6924 | 0.8459 | 0.6924 | 0.7402 |
| Logistic Regression + classweight | 0.6840 | 0.8321 | 0.6840 | 0.7342 |
| MultinomialNB + SampleWeight | 0.6381 | 0.8780 | 0.6381 | 0.6979 |
| DecisionTree + classweight | 0.6802 | 0.7095 | 0.6802 | 0.6880 |
| BernoulliNB + SampleWeight | 0.5477 | 0.7951 | 0.5477 | 0.5995 |
| LightGBM + classweight | 0.5187 | 0.7232 | 0.5187 | 0.5852 |
| XGB + SampleWeight | 0.4411 | 0.6937 | 0.4411 | 0.5140 |

**Figure 7: Results from short description with class weight technique and lemmatization**

**Base Model**

| Input | Imbalance Technique | Best Algorithm | F1-score | Performance Increase |
|---|---|---|---|---|
| Short Description | Random Over Sampling | TF-IDF + SVM | 0.7912 | - |

**Improve Imbalance Technique**

| Input | Imbalance Technique | Best Algorithm | F1-score | Performance Increase |
|---|---|---|---|---|
| Short Description | Class Weight | TF-IDF + SVM | 0.7974 | 0.0062 |

**Improve Text Normalization using Lemmatization**

| Input | Imbalance Technique | Best Algorithm | F1-score | Performance Increase |
|---|---|---|---|---|
| Short Description | Class Weight | TF-IDF + SVM | 0.7977 | 0.0003 |

**Figure 8: Summary of Experiment 1: finding the best class imbalance technique**

## 5.2 Experiment 2: Finding the best input features

From the previous experiment, the best class imbalance technique is to use a class weight technique, and the lemmatization technique also helps improving the performance for SVM model. Several experiments were explored using different input features and machine learning models to confirm that SVM gives the best performance. The results in Table 3 showed that SVM outperformed other models in terms of accuracy at 85.59% and F1-score at 85.71% using a class weight technique and lemmatization. Hybrid input features can increase the accuracy and F1-score. Most classical machine learning models showed that Hybrid 5 input feature, i.e., Short description, Description, Business unit, Sub business unit, and Company code, gave the best accuracy and F1-score when comparing to other input feature combinations. This is because the organizational contextual features can help the model to scope the selection of application and the F1-score is increased significantly from the baseline 1 input. However, Bernoulli Naïve Bayes model gave the lowest accuracy at 56.57% and F1-score at 61.45% for Hybrid 5 input feature and it is not suitable for IT Ticket classification.

| Input feature | Accuracy | Precision | Recall | F1-Score | F1-Score |
|---|---|---|---|---|---|

| combination | | | | | improvement |
|---|---|---|---|---|---|
| **Support Vector Machine (SVM)** | | | | | |
| **Baseline 1** | 0.7813 | 0.8356 | 0.7813 | 0.7974 | - |
| **Hybrid 2** | 0.8014 | 0.8345 | 0.8014 | 0.8095 | +1.21% |
| **Hybrid 3** | 0.8298 | 0.8500 | 0.8298 | 0.8420 | +4.46% |
| **Hybrid 4** | 0.8402 | 0.8522 | 0.8402 | 0.8420 | +4.46% |
| **Hybrid 5** | 0.8559 | 0.8665 | 0.8559 | **0.8571** | **+5.97%** |
| **Logistic Regression (LR)** | | | | | |
| **Baseline 1** | 0.6851 | 0.8329 | 0.6851 | 0.7353 | - |
| **Hybrid 2** | 0.7021 | 0.8004 | 0.7021 | 0.7219 | -1.34% |
| **Hybrid 3** | 0.7171 | 0.8017 | 0.7171 | 0.7353 | 0.00% |
| **Hybrid 4** | 0.7262 | 0.8053 | 0.7262 | 0.7418 | 0.65% |
| **Hybrid 5** | 0.7453 | 0.8129 | 0.7453 | **0.7568** | **2.15%** |
| **Multinomial Naive Bayes (MNB)** | | | | | |
| **Baseline 1** | 0.6382 | 0.8782 | 0.6382 | 0.6981 | - |
| **Hybrid 2** | 0.6376 | 0.8764 | 0.6376 | 0.6964 | -0.17% |
| **Hybrid 3** | 0.6369 | 0.876 | 0.6369 | 0.6917 | -0.64% |
| **Hybrid 4** | 0.6389 | 0.8699 | 0.6389 | 0.6884 | -0.97% |
| **Hybrid 5** | 0.6502 | 0.8609 | 0.6502 | **0.695** | **-0.31%** |
| **Bernoulli Naive Bayes (BNB)** | | | | | |
| **Baseline 1** | 0.5482 | 0.7943 | 0.5482 | 0.6001 | - |
| **Hybrid 2** | 0.531 | 0.7278 | 0.531 | 0.5712 | -2.89% |
| **Hybrid 3** | 0.5602 | 0.7563 | 0.5602 | 0.6022 | 0.21% |
| **Hybrid 4** | 0.5621 | 0.7639 | 0.5621 | 0.6052 | 0.51% |
| **Hybrid 5** | 0.5657 | 0.7865 | 0.5657 | **0.6145** | **1.44%** |
| **Stochastic Gradient Descent (SGD)** | | | | | |
| **Baseline 1** | 0.6935 | 0.8457 | 0.6935 | 0.7412 | - |
| **Hybrid 2** | 0.6963 | 0.8179 | 0.6963 | 0.7203 | -2.09% |
| **Hybrid 3** | 0.7081 | 0.8199 | 0.7081 | 0.7297 | -1.15% |
| **Hybrid 4** | 0.7085 | 0.8174 | 0.7085 | 0.7272 | -1.40% |
| **Hybrid 5** | 0.7221 | 0.8175 | 0.7221 | **0.7342** | **-0.70%** |

**Table 3: Hybrid input features performance comparison with different machine learning models using class weight technique and lemmatization**

| Input feature combination | Accuracy | Precision | Recall | F1-Score | F1-Score improvement |
|---|---|---|---|---|---|
| **Decision Tree (DT)** | | | | | |
| **Baseline 1** | 0.6814 | 0.7106 | 0.6814 | 0.6898 | - |
| **Hybrid 2** | 0.676 | 0.681 | 0.676 | 0.6726 | -1.72% |
| **Hybrid 3** | 0.5874 | 0.6401 | 0.5874 | 0.5868 | -10.30% |
| **Hybrid 4** | 0.6061 | 0.6681 | 0.6061 | 0.6077 | -8.21% |
| **Hybrid 5** | 0.6312 | 0.6846 | 0.6312 | **0.6315** | **-5.83%** |
| **Extreme Gradient Boosting (XGB)** | | | | | |
| **Baseline 1** | 0.4408 | 0.6934 | 0.4408 | 0.514 | - |
| **Hybrid 2** | 0.4607 | 0.7179 | 0.4607 | 0.536 | 2.20% |
| **Hybrid 3** | 0.6042 | 0.6797 | 0.6042 | 0.6098 | 9.58% |
| **Hybrid 4** | 0.6384 | 0.7014 | 0.6384 | 0.6442 | 13.02% |
| **Hybrid 5** | 0.6673 | 0.7192 | 0.6673 | **0.6707** | **15.67%** |
| **Light Gradient-Boosting Machine (LightGBM)** | | | | | |
| **Baseline 1** | 0.5175 | 0.7307 | 0.5175 | 0.5869 | - |
| **Hybrid 2** | 0.5662 | 0.7378 | 0.5662 | 0.6257 | 3.88% |
| **Hybrid 3** | 0.6348 | 0.6981 | 0.6348 | 0.6437 | 5.68% |
| **Hybrid 4** | 0.6541 | 0.7153 | 0.6541 | 0.6614 | 7.45% |
| **Hybrid 5** | 0.6781 | 0.734 | 0.6781 | **0.6843** | **9.74%** |

**Table 3: Hybrid input features performance comparison with different machine learning models using class weight technique and lemmatization (cont.)**

# 6 Conclusion and Future work

IT ticket classification problem is a crucial research topic for IT outsourcing company to increase the business process performance within Service Level Agreement (SLA). This paper proposed to use machine learning models, i.e. Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Logistic Regression (LR), Multinomial Naïve Bayes (MNB), Decision Tree (DT), Bernoulli Naïve Bayes (BNB), Light Gradient-Boosting Machine (LightGBM), and Extreme Gradient Boosting (XGB) to classify IT tickets. The dataset is from an IT outsourcing company with 20,586 records to be classified into 142 applications. The performance metrics are accuracy, precision, recall, and F1-score. Two sets of experiments, i.e. Experiment 1: finding the best class imbalance technique and Experiment 2: finding the best input features, were conducted. From the experiments using a class weight with lemmatization, and all 6 input features, SVM gives the best F1-score at 85.71% and the features are short description, description, business unit, sub business unit, and company code.

The classical machine learning models were thoroughly explored in this paper. For the future work, Recurrent Neural Network (RNN) and Bidirectional Encoder Representations from Transformers (BERT), which are modern machine learning techniques, will be used as classification techniques to enhance the performance to IT ticket classification problem.

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